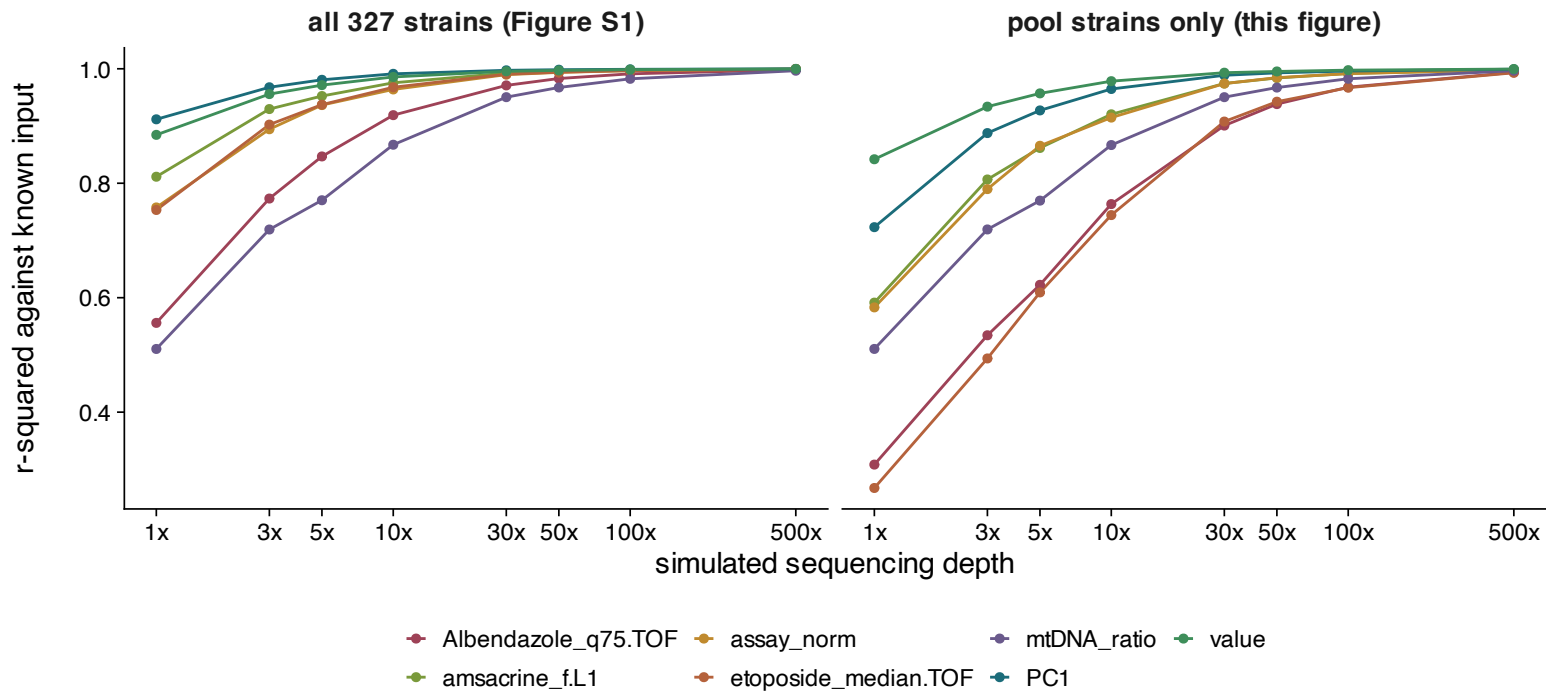


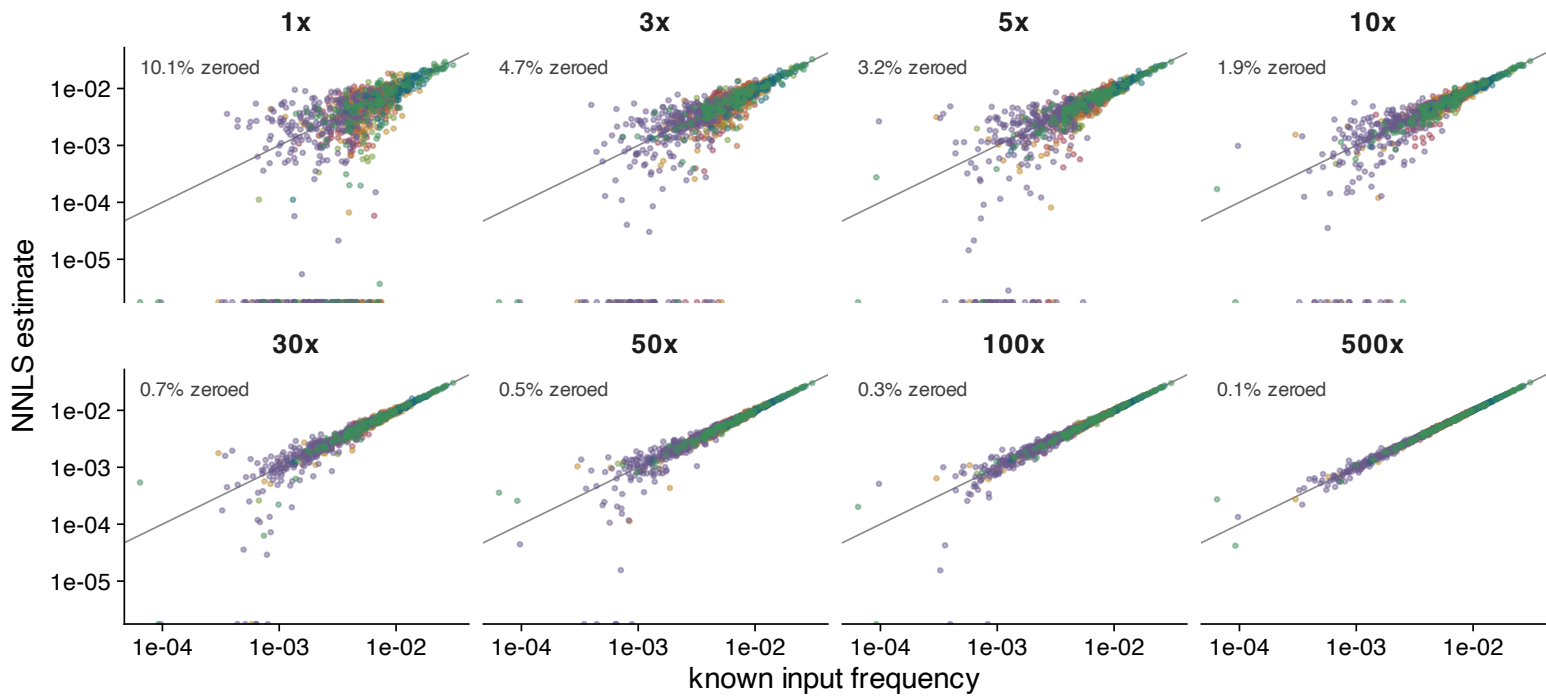
A Recovery against depth, with and without the absent strains

Mean over 10 replicates. Dropping the strains that were never in the pool costs 0.19 of r-squared at 1x and 0.07 at 10x.



B Estimate against truth, pool strains only

Replicate 1. Both axes log, which Figure S1 cannot do while the absent strains are in: their true value is exactly 0. The annotation counts strains that WERE in the pool and were returned as exactly zero. They cannot be drawn on a log axis, and they are the failure mode that matters – a zeroed strain contributes no phenotype. At 10x, 87 of 327 strains are zeroed in at least one trait; it tracks both abundance ($\rho = -0.20$) and privateness ($\rho = -0.25$).



C Recovery against pool size, suggestive only

Traits labelled at 1x. Recovery does fall as the pool grows, at every depth (Spearman -0.79 to -0.64), but with seven traits that is $p = 0.09$ at most depths and NOT significant. The one depth clearing 0.05 is one of eight correlated looks at the same seven points. Dispersion of the input frequencies also moves r-squared and the other way (ρ about $+0.64$); the two happen to be rank-independent here ($\rho = +0.000$), so they at least do not have to be disentangled. A series the simulation happened to contain, not one it was built to provide.

